

Siddharth Vaddem

623-273-8224 | svaddem@asu.edu | [linkedin.com/in/siddharthvaddem](https://www.linkedin.com/in/siddharthvaddem) | github.com/siddharthvaddem

EDUCATION

Master of Science in Computer Science, Arizona State University

Tempe, AZ

Cloud Computing, Algorithms, Distributed Systems, Databases — GPA: 4.00

August 2023 – May 2025

EXPERIENCE

Software Engineer Intern

June 2025 – August 2025

Modern Treasury (YC S18)

San Francisco, CA

- Redesigned enterprise billing ETL pipeline, migrating from Ruby on Rails to Dagster, dbt, and Snowflake; simplified the legacy codebase, cut execution time by 70%, and prevented main database resource contention while enabling usage-based billing for new product verticals.
- Enabled cell-based cost attribution across multi-region deployments via Terraform and added smoke tests for canary releases using Ruby, RSpec, TypeScript, and Buildkite.
- Built Sidekiq middleware to block oversized batches, preventing Redis saturation and circuit breaker triggers, eliminating multi-service outages and improving system reliability across fleets.
- Secured Spark-Snowflake ETL pipelines with key-pair authentication, supporting daily processing of 2.5M+ unreconciled transactions for ML-driven financial inference.

Software Engineer Intern

June 2024 – May 2025

Centso Investing

Blacksburg, VA

- Developed key financial workflows (user-driven basket generation, KYC/AML compliant onboarding) using React, Django, REST APIs, PostgreSQL, Auth0 and Azure.
- Reduced daily market data processing time from 1.5 hours to 16 minutes, enabling faster computation for investment style box visualizations in user-generated baskets.
- Optimized quant algorithm service response time by 25% and integrated it into serverless functions for asynchronous processing and high-throughput workloads using multiprocessing, caching, and vectorized computations.
- Automated financial securities metadata processing and SEC filings, saving 1000+ hours quarterly using Selenium, psychopg, Azure Blob Storage, and Runbooks.
- Enhanced email deliverability, reducing hard bounce rates by 15% through a serverless email management solution leveraging Azure Communication Services and automated bounce/complaint tracking.

SDE Intern

June 2021 – September 2021

Mystry Inc

Remote

- Engineered core components of the e-commerce workflow using React and Firebase, as well as an internal inventory management tool, enhancing user interaction and inventory tracking.
- Developed a Pokémon card-inspired sneaker recommendation game, using a Tinder-style swiping interface, leading to a 2,900% increase in user engagement and enhancing data collection for the ML team.

PROJECTS

EdgeSight | AWS (EC2, S3, SQS, Greengrass, Lambda, IoT Core), Flask

March 2025 – April 2025

- Engineered a fault-tolerant, event-driven real-time face recognition platform with custom EC2 autoscaler; redesigned architecture using AWS Greengrass, IoT Core, and Lambda for low-latency edge inference.

SketchX | Tensorflow, OpenCV, GANs, Colab

December 2022 – April 2023

- Developed a GAN model to convert sketches into realistic portraits for law enforcement suspect identification.
- Used U-Net and PatchGAN architecture, applying transfer learning to cut FID score by 36% and enhance realism.

Huddl Collab | React, Node.js, Express, WebRTC, CRDT

May 2022 – July 2022

- Developed a real-time collaboration tool with document editing, audio, and video communication using WebRTC.
- Implemented CRDT for conflict-free state synchronization using Yjs, ensuring consistency in concurrent edits.

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, C++, SQL, HTML, CSS.

Frameworks/Libraries: React, Node.js, Express, Django, FastAPI, D3.js, Firebase, MongoDB, Selenium, Pandas.

Developer Tools: Git, Docker, Kubernetes, PostgreSQL, Redis, AWS (S3, EC2, SQS, SES, Lambda), Azure, Auth0, Terraform, YAML, Nginx, Linux/Unix.